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Feasibility for non-destructive discrimination of natural and beryllium-diffused sapphires using Raman spectroscopy

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ABSTRACT

Raman spectroscopy based non-destructive discrimination between natural and beryllium-diffused (Be-diffused) sapphires has been attempted. The initial examination of Raman image acquired on a sapphire revealed that microscopic structural and compositional heterogeneity was apparent in the sample, so acquisition of spectra able to represent a whole body of sapphire rather than a localized area was necessary for a reliable discrimination. For this purpose, a wide area illumination (WAI) scheme (illumination area: 28.3 mm^2) providing a large sampling volume was employed to collect representative Raman spectra of sapphires. Upon the diffusion of Be into a sapphire, the band shift originated from varied lattice structure by substitution of Be at cation sites was observed and utilized as a valuable spectral signature for the discrimination. In the domain of principal component (PC) scores, the groups of natural and Be-diffused sapphires were identifiable with minor overlapping and the cross-validated discrimination error was 7.3% when *k*-Nearest Neighbor (*k*-NN) was used as a classifier.
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1. Introduction

Sapphire, one of most popular gemstones, is basically aluminum oxide (Al_2O_3) incorporating trace amounts of metals such as Mg^{2+}, Fe^{2+}, Fe^{3+}, Ti^{4+}, Cr^{3+} and V^{3+}. As generally known, Al_2O_3 has several structural phases such as water-free, α-Al_2O_3 and γ-Al_2O_3 [1,2]. Various colors of sapphires are originated from the charge-compensation mechanisms (intra-atomic absorption) or inter-valence charge transfer absorption via the conduction and valence band depends on incorporated metals [3–8]. Recently, several efforts have been accomplished to artificially alter the color of a sapphire to increase its value as a jewel and one of most common approaches is diffusion of beryllium (Be) into a sapphire at highly elevated temperatures over 1800 °C [5,9–12]. The addition of Be can form discrete energy levels by causing the electron to be transferred to the Be^{2+}-hole acceptor (inter-valence charge transfer) [10,13]. Either natural or Be-diffused sapphires possessing the same color are available in market, so the identification of their origins especially based on a non-destructive analysis is greatly demanding for fair value evaluation.

Here, we have attempted to differentiate yellow natural and Be-diffused sapphires using Raman spectroscopy since it provides sensitive spectral information pertinent to varied lattice structure without any destructive sampling, thereby an optimal choice for the non-destructive evaluation of sapphires. The colors of both samples are nearly identical as shown in Fig. 1, so simple visual examination is unable to distinguish them when they are mixed each other. To initially investigate microscopic structural and compositional characteristics of natural and Be-diffused sapphires, Raman mapping was performed on a polished surface of both samples and their microscopic structural and compositional homogeneity was examined. If a sample is internally inhomogeneous, it is critically necessary to acquire representative spectra of a whole sapphire rather than a localized area for a reliable spectroscopic discrimination. To acquire representative Raman spectra, a wide area illumination (WAI) scheme [14–16] capable of covering a large sample area (28.3 mm^2, laser illumination diameter: 6 mm) was used. Then, a spectral signature valuable for the discrimination between natural and Be-diffused sapphires was searched. Finally, principal component analysis (PCA) [17–22] was used to represent the spectral feature in a score domain, and then both linear discriminant analysis (LDA) [17,19,20,22] and *k*-nearest neighbor (*k*-NN) [20,23] were applied to determine the samples as either natural or Be-diffused sapphires.

2. Experimental

The picture of 20 natural and 21 Be-diffused sapphires employed in this study is shown in Fig. 1. The samples were obtained from a reliable trader in Seoul, Korea and their sizes ranged from

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Fig. 1. Picture of yellow natural and Be-diffused sapphires employed in this study.

3 to 7 mm approximately. The colors of all sapphires are similar, so identifying each sapphire as either natural or Be-diffused by just naked-eye examination is practically unachievable.

The WAI scheme (PhAT probe, Kaiser Optical Systems, USA) provided a circular laser (785 nm) illumination with a diameter of 6 mm to cover a large sample area (28.3 mm^2) [14–16]. A sapphire was simply positioned at the focal point of laser illumination for spectral collection. Each Raman spectrum (resolution: 4 cm^{-1}) was acquired by an accumulation of 8 scans with 5-second exposure at each scan. Further, Raman mapping was performed on a polished surface of a sectioned sapphire. The sectioned sample was positioned on a microscope stage connected with Raman spectrometer (Kaiser Optical Systems) and laser radiation was focused using an objective lens (10 × 0.25 numerical aperture (NA)) to collect Raman spectra. The laser spot size was approximately 50 μm in diameter. Baseline correction of raw spectra, intensity normalization, PCA, LDA and *k*-NN were performed using Matlab version 7.0 (The Math-Works Inc., MA, USA).

3. Results and discussion

3.1. Raman mapping of a natural and Be-diffused sapphire

To probe microscopic structural and compositional variation of a sapphire by Be-diffusion, Raman mapping was performed. For this purpose, one sapphire from each group was selected and its surface was polished for the mapping. The sizes of two selected sapphires were similar (approximately 5.2 × 3.3 mm), while the Be-diffused sapphire was minutely larger. The mapping intervals for the natural sapphire were 258 and 135 μm in *x* and *y* directions, respectively, and 258 and 168 μm for the Be-diffused sapphire. In both mappings, a total of 400 (20 × 20) spectra were collected.

To initially examine overall spectral feature of natural and Be-diffused sapphires, all of 400 spectra acquired at each mapping was averaged and the corresponding average spectra were

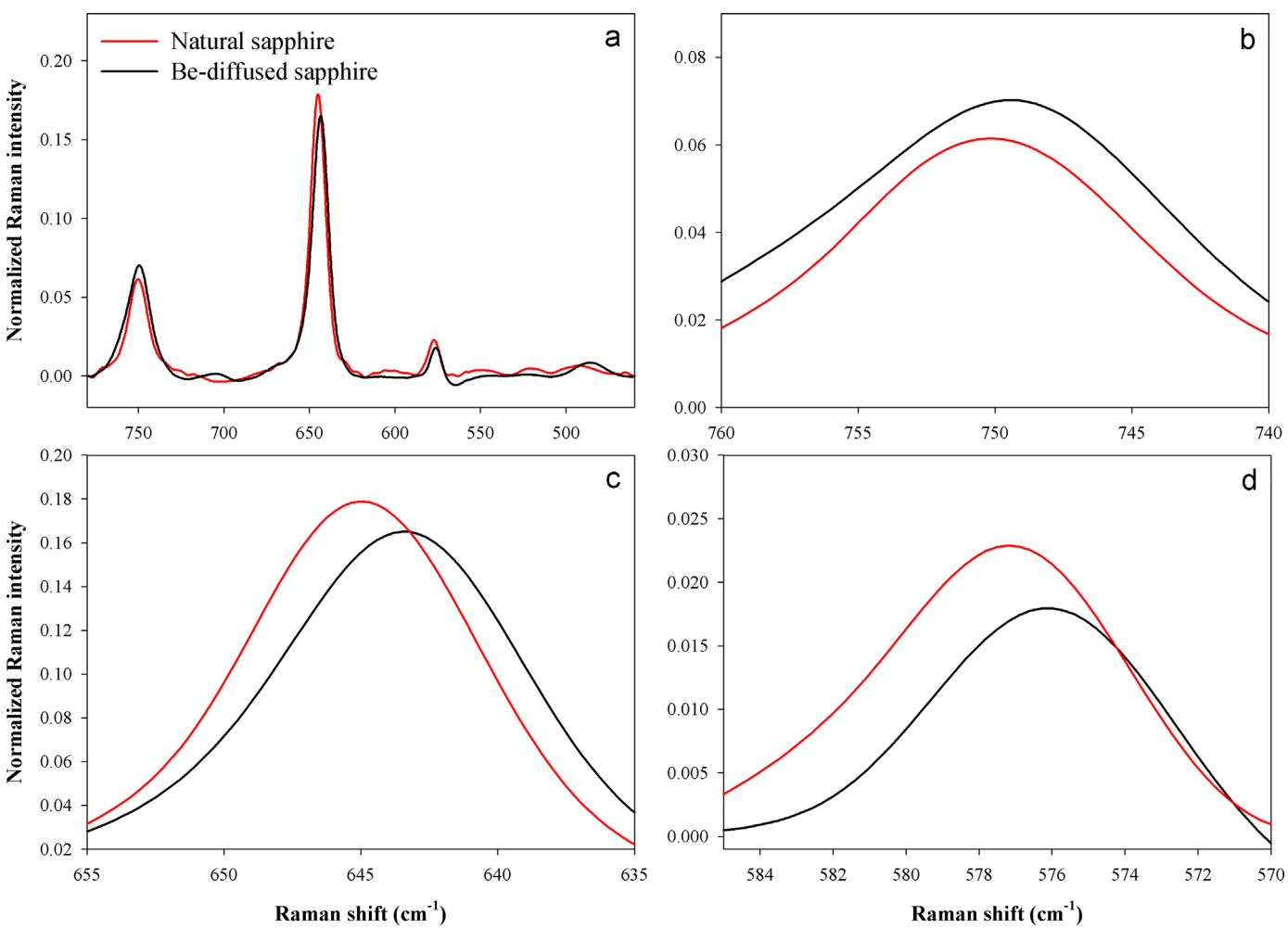

Fig. 2. Spectra of natural (red) and Be-diffused (black) sapphires acquired by averaging all of 400 spectra acquired at each mapping (a). For detail observation, the 750, 645 and 577 cm^{-1} bands are individually magnified in Fig. 2(b), (c) and (d), respectively.

compared as shown in Fig. 2(a). Red and black colors indicate natural and Be-diffused sapphires, respectively. Before the averaging, the baselines of raw spectra were linearly corrected at 780, 600 and 460 cm^{-1}, and the baseline-corrected spectra were divided by the corresponding peak area under the 780–460 cm^{-1} range for normalization. For detail observation, the 750, 645 and 577 cm^{-1} bands are individually magnified in Fig. 2(b), (c) and (d), respectively. The 750, 645 and 577 cm^{-1} bands correspond to vibrations of Al_2O_3 in E_g (external), A_{1g} (internal) and E_g (internal) modes, respectively [24].

The noticeable observation in these bands is the shift of peak position to the lower wavenumber as a result of Be-diffusion. The magnitude of shift varies from 1 to 3 cm^{-1} approximately. In a meanwhile, no appearance of new bands or disappearance of existing bands is observed for the Be-diffused sapphire, thereby informing no substantial compositional change by Be-diffusion. Therefore, the band shift is most probably originated from structural disorder associated with dissolutions of Be, and the valence changes also induce charge-disorder which causes variation in the bond lengths [13]. The substitution of cations does not change a space group, however it slightly modifies the lattice parameters, thereby resulting in the minute shift of Raman band.

Then, microscopic spectral variation induced by Be-diffusion was investigated using Raman images as shown in Fig. 3. Fig. 3 (a) presents Raman image of the natural sapphire constructed by the position of 645 cm^{-1} band. Different colors are used to indicate the peak positions from 647 to 643 cm^{-1} (refer to the color scheme in the figure). As shown, the color variation in the image is apparent, so indicative of microscopic heterogeneity in the lattice structure. Fig. 3(b) shows Raman spectra collected at three different spots with the greater color contrast as indicated by the circles in the image. The peak positions as well as shapes of the spectra vary fairly. Especially the red band (spectrum #1) seems to contain two different peaks, so informing also possible compositional inhomogeneity. While, the colors of the image are completely different for the Be-diffused sapphire as shown in Fig. 3(c). Blue color is nearly uniform over the sample due to the consistency in the peak position. Raman spectra collected at three randomly selected spots marked by the circles on the image are also shown Fig. 3(d). The band positions and shapes are nearly similar with each other. This is attributed to internal recrystallization and artificial healing of fractures induced by the diffusion of Be into a sapphire at the extreme high temperature. In addition, internal voids that contain water and a bubble of carbon dioxide cannot survive at this condition [10].

Fig. 4(a) and (c) shows Raman images of the natural and Be-diffused sapphire, respectively, constructed using the peak area in the 535–510 cm^{-1} range. In the image of natural sapphire, the localized areas with red/orange color are observed; while, these colors are unobservable in the image of Be-diffused sapphire. Fig. 4 (b) shows Raman spectra (780 – 460 cm^{-1} range) acquired at three different spots marked by the circles in the image of natural sapphire. Spectra collected at the spot 1 and 2 show strong Raman intensity at 520 cm^{-1}, while absent at the spot 3. In contrast, the 520 cm^{-1} peak completely disappears as shown in Fig. 4(d), which presenting spectra measured at 3 randomly selected spots on the

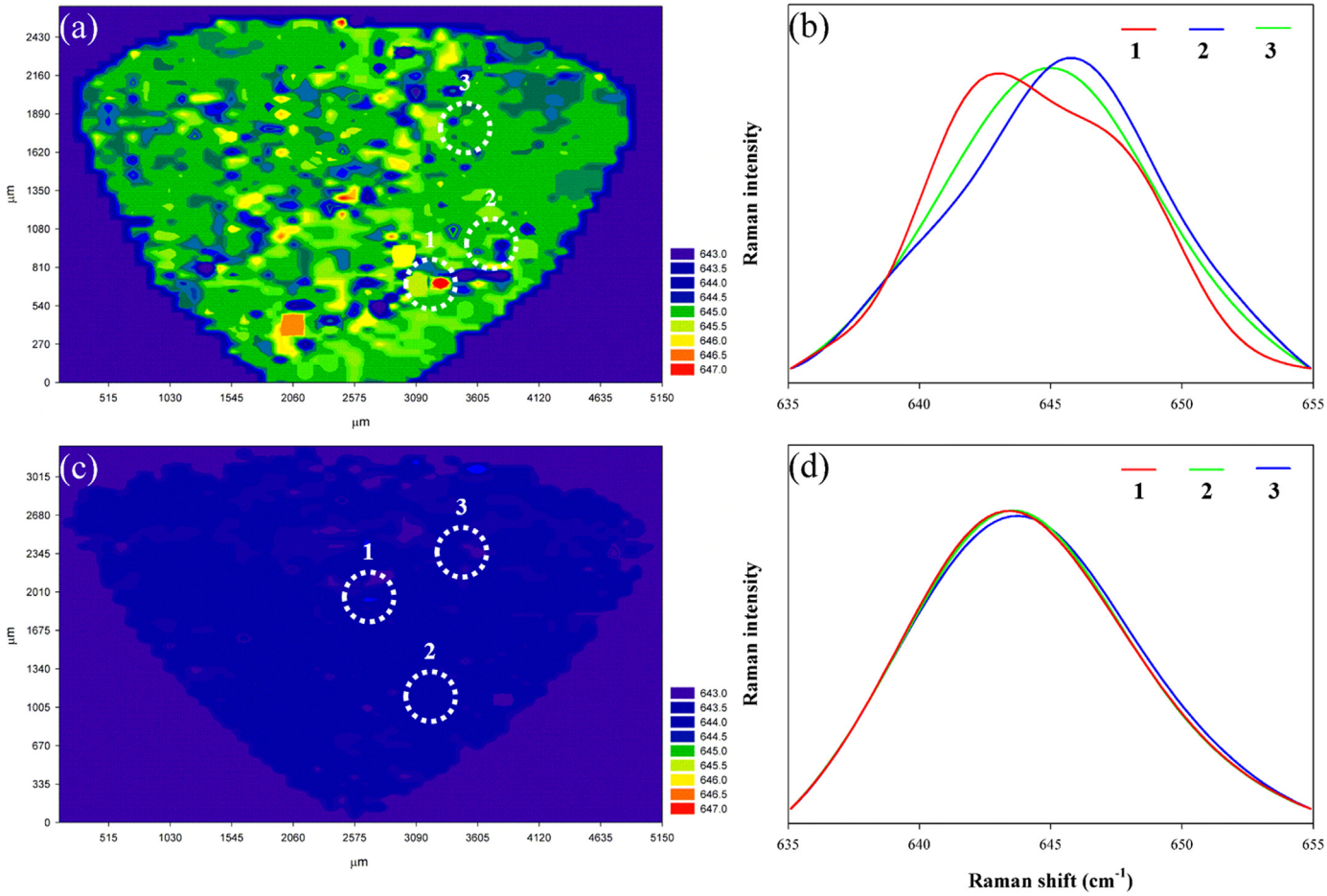

Fig. 3. Raman image of the natural (a) and Be-diffused (c) sapphire constructed by the position of 645 cm^{-1} band. Different colors are used to indicate the peak positions from 647 to 643 cm^{-1} (designated as red to blue, respectively). Raman spectra collected at three different spots indicated by the circles on the natural (b) and Be-diffused (d) sapphire are also shown.

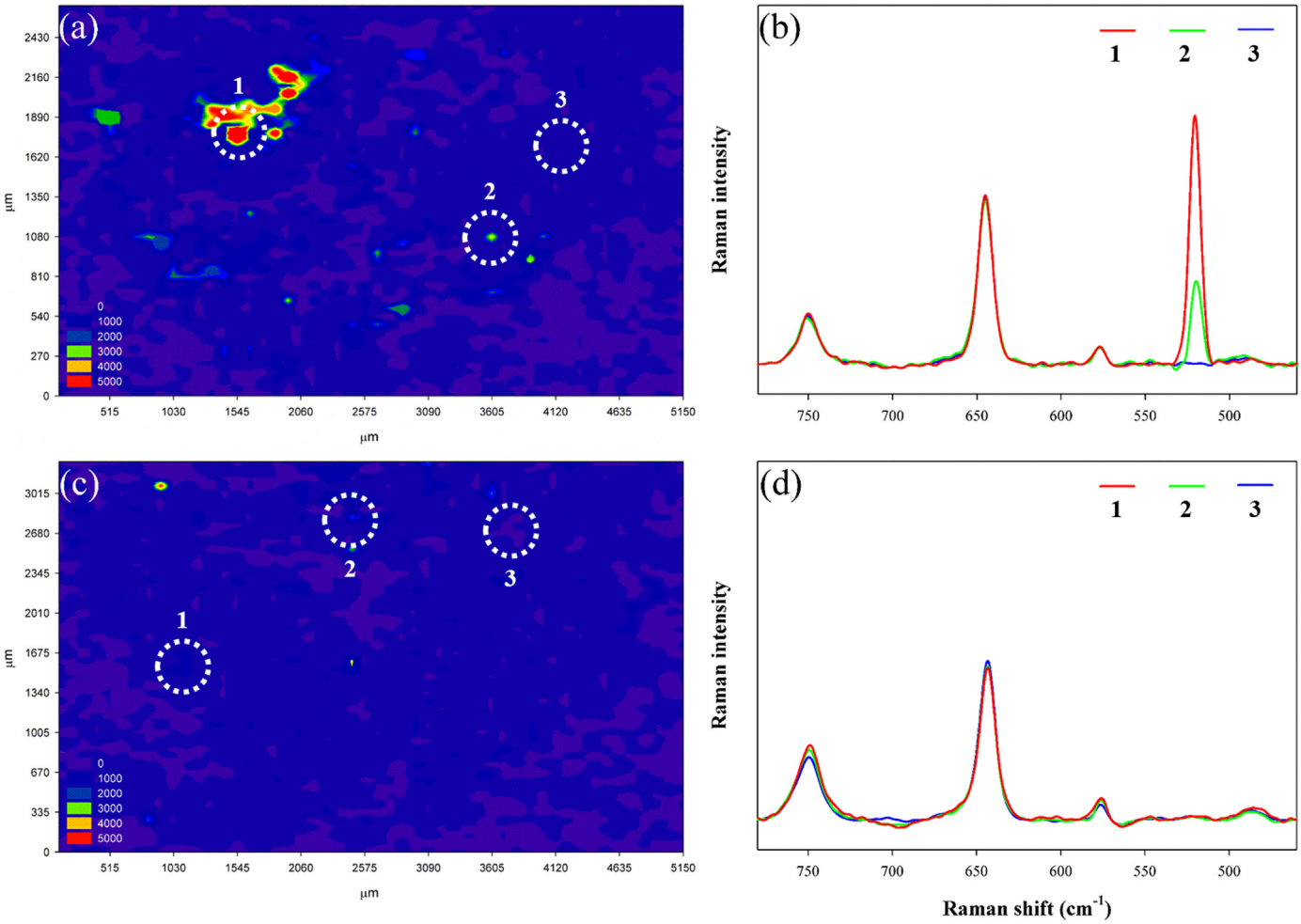

Fig. 4. Raman images constructed using the peak area in the 535–510 cm^{-1} range for natural (a) and Be-diffused (c) sapphire. Raman spectra collected at three different spots indicated by the circles on the natural (b) and Be-diffused (d) sapphire are also shown.

sample. This result illustrates that the natural inclusions are eliminated by such as incineration during Be-diffusion performed at temperature over 1800 °C [10].

The Raman mapping images in Figs. 3 and 4 inform that the internal structural heterogeneity and natural inclusions are present in a natural sapphire. In a meanwhile, the structure of a Be-diffused sapphire becomes uniform by internal recrystallization and removal of natural inclusions. Since internal heterogeneity in a natural sapphire is apparent, a spectrum obtained at only a localized area could be improper to represent a whole sample and possibly degrade discrimination accuracy. Therefore, acquisition of sample-representative Raman spectra of sapphires such as by covering a large sample volume is essentially demanding for accurate evaluation.

3.2. Discrimination between natural and Be-diffused sapphires

As described earlier, the WAI scheme was used for Raman spectral collection to cover nearly a whole sapphire body for the representative sampling. Although some sapphires are larger than the size of illumination diameter (6 mm), laser radiation propagates throughout a sample by internal refraction and reflection, and possibly interacts with a whole sample. Therefore, the scheme still allows acquisition of sample-representative Raman spectra of sapphires slightly larger than 6 mm. Relative standard deviations (RSDs) among 6 replicate spectra (normalized spectra in the 780–460 cm^{-1} range) collected by randomly repositioning a sample at each spectral collection were calculated from three sapphires of small, middle and large sizes. Average RSD was 0.4% (0.7, 0.4 and 0.2% for the small, middle and large sapphires, respectively), indicating the collection of reproducible and representative spectra of a sapphire using the WAI scheme.

Fig. 5(a) shows average Raman spectra acquired from the spectra of all natural (red) and Be-diffused sapphires (black) in the 780–460 cm^{-1} range and the 750, 645 and 577 cm^{-1} bands are separately magnified in Fig. 5(b), (d) and (f), respectively. All of individual spectra were normalized before the averaging as described earlier. Also, the 750, 645 and 577 cm^{-1} peaks of all the samples before the averaging are shown in Fig. 5(c), (e) and (g), respectively.

In the average spectrum of Be-diffused sapphires, the positions of three major bands shift to the lower wavenumber as similarly observed in Fig. 2. The altered lattice structure by the inclusion of Be is again responsible for the band shift. When the individual peaks of all samples are compared together, the spectral variations such as changes in peak intensity as well as position are greater for the Be-diffused sapphires compared to the natural sapphires. This is presumably from the use of diverse inexpensive sapphires for Be-diffusion to upgrade their commercial value, so the larger spectral variation is resultant. In overall, the difference in the spectral features between the natural and Be-diffused sapphires, especially the shift of band by Be-diffusion, should be a valuable source for their spectroscopic discrimination.

To simply compare the spectral features of samples together,

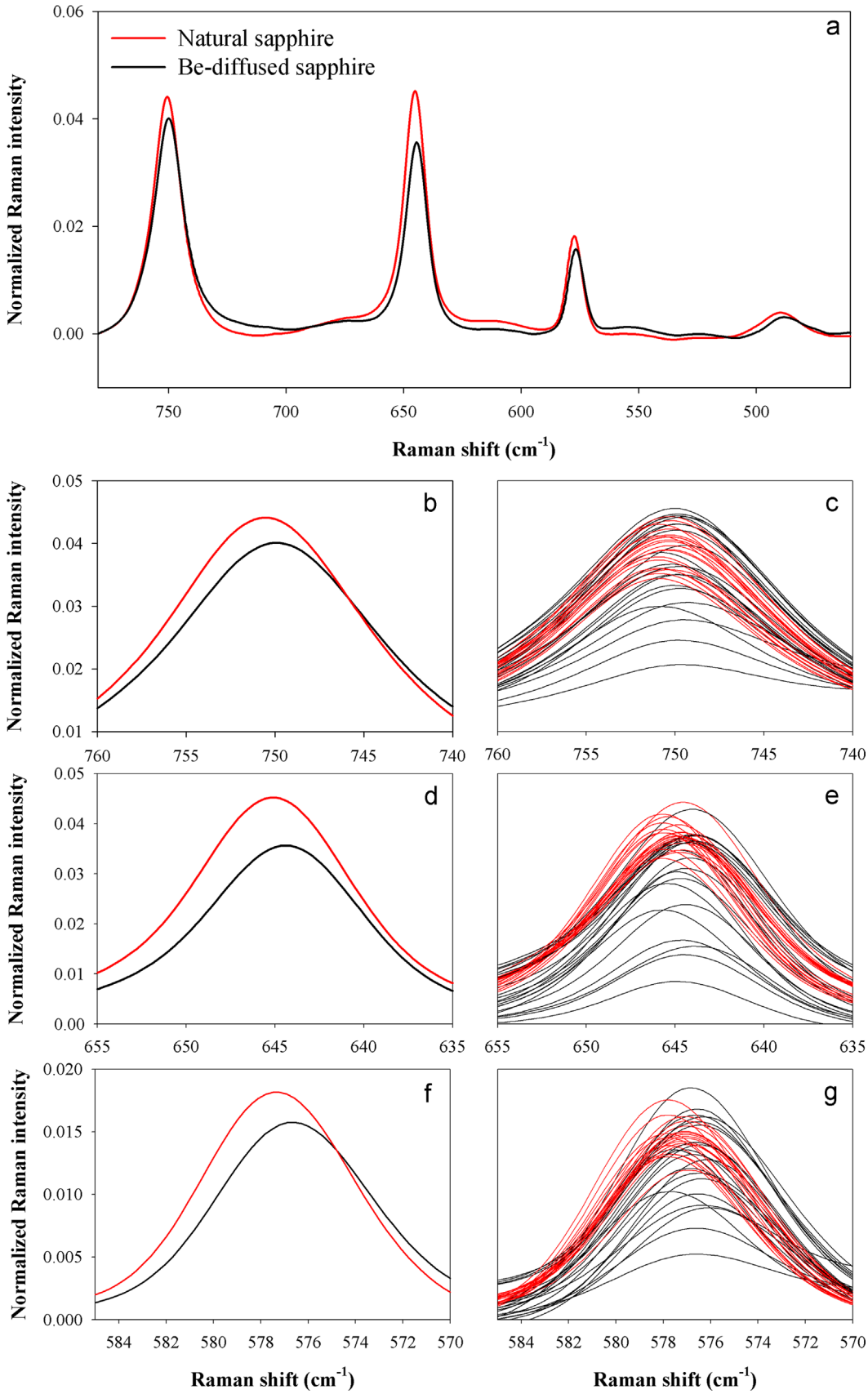

Fig. 5. Average Raman spectra acquired from the spectra of all natural (red) and Be-diffused sapphires (black) in the 780–460 cm^{-1} range (a). The 750 (b), 645 (d) and 577 cm^{-1} (f) bands are separately magnified. Also, the 750 (c), 645 (e) and 577 cm^{-1} (g) bands of all sapphire samples before the averaging are shown.

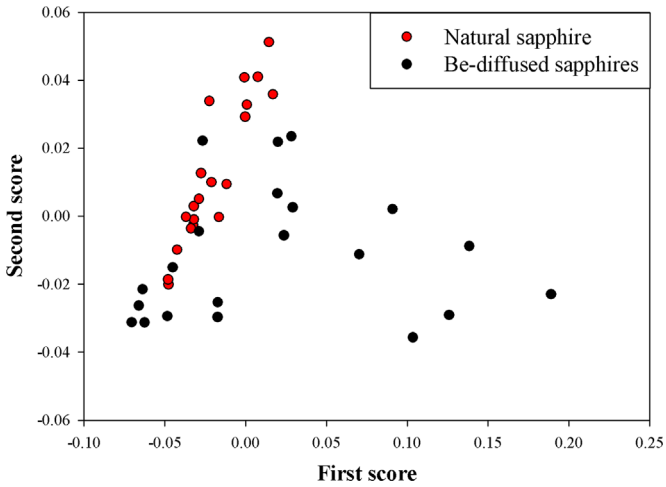

Fig. 6. Scatter plot constructed by the first and second scores. The first and second factors describe 77.7% and 12.4% of the total spectral variation.

PCA was performed using the spectra in the 780–460 cm^{-1} range and the resulting scores were examined. Fig. 6 show the scatter plot constructed by the first and second scores. The first and second factors described 77.7% and 12.4% of total spectral variation present in the dataset. Red and black scores correspond to natural and Be-diffused sapphires, respectively. The score groups corresponding natural and Be-diffused sapphires are apparent and fairly discernible with each other, although some scores from both groups are slightly inter-crossed. As expected, the score distribution of Be-diffused sapphires is wider due to the larger spectral variation as observed in Fig. 5.

Both LDA and *k*-NN were employed to distinguish the samples into two groups. As known, LDA [17,19,20,22] classifies objects into one of two groups by maximizing the ratio of between-class variance to within-class variance; while, *k*-NN [20,23] classifies unknown samples based on their similarity (or similarity measure) with samples in a training set. A leave-one-out cross-validation was used to estimate the accuracy of discrimination using the first and second scores as shown. The resulting discrimination errors were 9.8% and 7.3% when LDA and *k*-NN ($k=7$) were employed, respectively. The more accurate discrimination was achieved with the use of *k*-NN.

4. Conclusion

The altered Raman spectral feature such as the band shift occurred from the substitution of Be at cation sites in a sapphire was the origin for spectral discrimination between natural and Be-diffused sapphires. The WAI scheme was handy to collect representative spectra of sapphires with different sizes and shapes. We have mainly demonstrated a feasibility study here for the discrimination, so expansion of dataset by incorporating more natural sapphire samples with diverse origins and Be-diffused samples could be a prerequisite to make the proposed method market-acceptable. The preparation of a statistically valid dataset incorporating a large number of diverse sapphire samples is under way. A non-destructive analytical method sampling is especially demanding in analysis of gemstones. In this sense, the WAI scheme is easy and effective to collect representative Raman spectra for samples with different shapes, thereby possibly useful for analysis other jewels such as ruby and tourmaline.

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